Python Programming Lab Assignments

List of Problem Statements and Submission Dates

S.No	Problem Statement	Submission Date
6	**Student Marks Processing using Nested Lists** Write a Python program to create a list and demonstrate list operations. Declare a nested list `students_marks` to store the details of 5 students, where each inner list holds the student's name followed by the marks in 5 subjects. Structure: `students_marks = [[], [], [], [], []]` Example: `student = ['Sachin', 50, 55, 67, 76, 89]` Implement the program using separate functions to: a) Read the name and 5 subject marks of each student from the keyboard and append the list to `students_marks`. b) Calculate the total marks of each student and append the total at the end of that student's list. c) Display the student name, subject marks and total of every student in a tabular format. Use list operations such as `append()`, indexing, slicing and `sum()` wherever applicable.	22/10/2026
7	**Bank Customer Record using Tuple and Nested Tuple** Write a Python program to demonstrate tuples and nested tuples by storing a list inside a tuple. Store the account number, account holder name and date of birth in a tuple, and store the email ID, contact number, address and balance in a list. The final structure should be: `(account_no, holder_name, dob, [email_id, contact_no, address, balance])` Read the data from the keyboard and display the account details in a tabular format. Also demonstrate that the tuple elements cannot be modified, while the balance inside the nested list can be updated.	22/10/2026

8	**Menu-Driven Bank Account Management**
	Write a menu-driven Python console application to process N bank accounts with Deposit, Withdraw and Show Accounts modules. Store each account in a list and all N accounts in a nested list:
	`accounts = [account_no, name, email_id, address, balance]`
	`bank_accounts = [[account_no, name, email_id, address, balance], ...]`
	Implement each module as a function:
	a) **Deposit**: search for the account number, add the amount and display the current balance.
	b) **Withdraw**: search for the account number, check for sufficient balance, deduct the amount and display the current balance.
	c) **Show Accounts**: display all the accounts in a tabular format.
		22/10/2026

S.No	Problem Statement	Submission Date
	Display "Account not found" for an invalid account number and reject zero or negative amounts. Show the menu again after each operation until the user chooses Exit. **Sample Output:** 1. Deposit 2. Withdraw 3. Show Accounts 4. Exit Enter your choice: 1 Enter account number: 1000 Account found... Please enter deposit amount: 600 Current balance: 2100 --- Enter your choice: 2 Enter account number: 2000 Enter withdraw amount: 500 Current balance: 1000 --- Enter your choice: 3 (All accounts displayed in a tabular format)	